

OS-Level Omnipresent Copilot Framework

Patterns and decision rules for balancing a global intent bar, contextual right sidebar copilot, inline assistance, and proactive activity rails across an enterprise desktop suite for bankers/advisors.

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One-line principle: The intent bar routes. The sidebar works. Inline assists. The rail nudges.

Why a framework first

An OS-level copilot must behave consistently across many apps, records, and workflows. A shared framework prevents the assistant from becoming a "junk drawer" UI, accelerates design decisions, and creates a stable contract for engineering (what opens where, when, and why).

Scope and assumptions

- The intent bar is OS-level and always available across the product suite (global entry point).
- Users operate in multiple apps (CRM, brokerage, cases, content/library, research) with frequent context switching.
- Copilot may not yet execute all actions end-to-end; the experience must support "plan + handoff" gracefully.
- Regulated environment: permission boundaries, audit trails, and confirmation gates are first-class.

1. The core model: Locality x Depth

Every copilot request can be classified on two axes. These axes determine which surface should lead and which pattern to use.

Axis A: Context locality (where the required information and action live)

- **Global (OS-level):** across apps and the book of business. Example: "Who should I talk to right now?"
- **App:** within a single application context. Example: "What routine account maintenance is outstanding in this app?"
- **Record:** tied to a specific client/account/case/document. Example: "What changed for this client since last week?"
- **Workflow/Form:** a specific transactional step. Example: "Help me complete this trade ticket / KYC review."

Axis B: Interaction depth (how many steps and how much review is required)

- **0-1 step:** lookup, navigate, quick answer.
- **2-4 steps:** refine, compare, choose, preview.
- **Multi-step / transactional:** creates, updates, sends, books, submits; requires validation and approvals.

Surface selection in one sentence

Locality decides *where* context comes from; depth decides *how much workspace* is needed. Global + shallow -> intent bar. Record + deep -> sidebar. Workflow + deep -> inline + sidebar.

2. Canonical copilot surfaces and responsibilities

Treat each surface as a product with a clear job. Do not overload one surface to cover all intents.

Surface	Primary role	Best for	Avoid
Omnipresent Intent Bar	Start + route + find + navigate	Global questions, entity search, app switching, quick answers	Long multi-turn work; transaction confirmations; dense evidence review
Right Sidebar Copilot (Workspace)	Multi-turn reasoning + artifacts + audit trail	Refinement, synthesis with sources, previews/approvals, guided flows, status tracking	Being the only entry point (creates friction for quick intent)
Inline Copilot (In-flow Assist)	Micro-assistance at the point of work	Form help, prefill, explanations, field-level suggestions	Full chat UI in forms; lengthy back-and-forth
Proactive Rail / Feed	Ambient intelligence + attention management	Urgent tasks, risks, opportunities, nudges with one-tap actions	Replacing explicit user intent capture (rail is not primary search)

3. Pattern library

Patterns are reusable interaction recipes. Each pattern names: entry point, rendering surface, expected output, and escalation behavior.

Command Palette Results

Use when: User intent is: find / open / navigate (client, account, case, doc, app).

UI: Intent bar -> results list with entity matches + quick actions.

Escalation: Escalate to sidebar if the user asks "why", needs comparison, or requests a follow-up.

Quick Answer Overlay

Use when: 0-1 step Q&A; with low risk; user stays in flow.

UI: Intent bar -> lightweight answer card with "View details".

Escalation: Escalate to sidebar if answer needs sources, joins data, or becomes multi-turn.

Deep Answer (Synthesis)

Use when: Cross-system or analytical question: risks/opps, KPI performance, book insights.

UI: Intent bar -> open sidebar with structured answer, sources, and recommended next actions.

Escalation: Default to sidebar to preserve context, traceability, and follow-up continuity.

Guided Action Flow

Use when: User wants to do something: send, book, submit, update, trade, create.

UI: Sidebar -> stepper (review, confirm, execute) with previews and guardrails.

Escalation: Never execute external actions without explicit confirmation gates.

Record Copilot (Pinned)

Use when: User is on a client/account/case record; needs "catch me up" and next best actions.

UI: Sidebar anchored to the current record (context chip is always visible).

Escalation: Offer one-click actions that launch guided flows or deep links.

Inline Assist (Form Micro-actions)

Use when: User is mid-transaction in a form/workflow (trade ticket, onboarding, KYC).

UI: Inline -> "prefill", "explain", and validation suggestions; full reasoning in sidebar.

Escalation: Inline should remain brief; move complex discussion to sidebar.

Proactive Card (Explain / Act / Snooze)

Use when: System detects urgent task, risk, or opportunity (excess cash, drift, overdue tasks).

UI: Rail -> card with three actions: Explain (sidebar), Act (guided flow), Snooze (work mgmt).

Escalation: Avoid pushing users into chat; keep proactive content skimmable.

Plan + Handoff (No Infra)

Use when: Copilot cannot execute end-to-end today; still needs to help users complete work.

UI: Sidebar -> plan, prerequisites, deep links into the right system, and a progress checklist.

Escalation: Be explicit about what is automated vs what the user must complete.

4. Locality x Depth mapping matrix (primary decision tool)

Pick locality (row) and depth (column). The cell recommends the leading surface and pattern.

Locality \ Depth	0-1 step (lookup/answer)	2-4 steps (refine/decide)	Multi-step / transactional
Global (OS-level)	Intent bar: Command Palette or Quick Answer. Output: results, shortcuts, brief answer.	Bar -> Sidebar: Deep Answer (synthesis). Output: sources + recommendations.	Sidebar: Guided Action or Plan + Handoff. Output: stepper, previews, confirmations.
App (single application)	Intent bar scoped to app: Quick Answer or palette.	Sidebar: Deep Answer within app context; comparisons.	Sidebar: Guided Action; Inline if form-based.
Record (client/account/ case/doc)	Sidebar preferred: Record Copilot "catch up". Bar can route into sidebar.	Sidebar: Deep Answer + recommendations + evidence.	Sidebar + Inline: Guided action with record context; strict gates.
Workflow / Form (transaction)	Inline: explain / prefill / validate.	Inline + Sidebar: refine details; preview; resolve exceptions.	Inline + Sidebar: Guided action with strict gates; confirmations.

Note: If a global request becomes record-specific (e.g., user selects a client from results), handoff into the sidebar and attach the selected record as explicit context.

5. Decision rules and escalation logic

5.1 Where does the interaction start?

- **Start in the intent bar** when scope is global/ambiguous, the user wants find/ask/navigate, or the expected interaction is 0-1 step.
- **Start in the sidebar** when the user is on a record, the task implies refinement, the output is an artifact, or the answer needs evidence/audit trail.
- **Start inline** when the user is inside a form/transaction and interruptions are costly.

5.2 Automatic escalation: Bar -> Sidebar

- The user intent is a "world-changing" verb: send, submit, approve, update, create, book, trade.
- The answer requires joining data across systems or explaining rationale/sources.
- The model must ask a follow-up question (signals multi-turn).
- The user requests: show sources, explain why, compare options, draft an artifact.

5.3 Proactive rules: Rail -> Sidebar/Flow

- Proactive items must be skimmable and actionable; default actions: Explain (sidebar), Act (guided flow), Snooze.
- Do not force users into chat to clear alerts; allow one-click resolution where possible.
- Use proactive only for: urgent tasks, high-confidence risks, and high-confidence opportunities.

5.4 Surface state model (recommended)

To avoid UI thrash across apps, use a consistent set of states for the copilot surfaces:

- **Collapsed**: icon only (sparkle).
- **Peek**: short overlay/panel for quick answers (dismissible).
- **Expanded**: full sidebar workspace with history.
- **Pinned**: stays open while navigating across apps/records; context chip updates.
- **Docked split**: optional mode where rail + copilot share the right side (tabs).

6. Trust, compliance, and enterprise guardrails (the trust contract)

In regulated enterprise work, copilot UX must make trust visible. The sidebar workspace is the primary home for these controls.

- **Scope disclosure:** always show what context is used (This screen / This record / This app / All apps).
- **Source attribution:** for synthesis, show evidence and provenance (data system, timestamp, link).
- **Permission awareness:** never leak across teams/books; signal when results are incomplete due to access.
- **Human-in-the-loop:** external communications and transactions require preview + explicit confirmation.
- **Audit trail:** log prompts, outputs, sources used, and user approvals (policy-dependent).
- **Honest capability:** when infra cannot execute, provide plan + handoff instead of pretending automation.

7. Interaction mechanics (OS-level across apps)

OS-level presence changes expectations: users will invoke copilot while switching apps and records. These mechanics keep behavior consistent and predictable.

- **Global shortcut:** a single keyboard gesture (e.g., Cmd/Ctrl+K) focuses the intent bar everywhere.
- **Scope chips:** one-tap toggles for context (Active window, Active record, This app, All apps, My book).
- **Context attachments:** bar results can attach entities (client/account/doc) into the sidebar as explicit context.
- **Handoff animation:** bar -> sidebar should feel like "continuing" the same request, not starting over.
- **Session continuity:** pinned sidebar retains thread across app navigation; thread header shows active context.
- **Deep links:** when executing requires another system, provide one-click links with a prefilled plan/checklist.

Appendix: Quick intent-to-pattern cheatsheet

User intent type	Default pattern	Surface
Find/open something (client/account/doc/app)	Command Palette Results	Intent bar
Ask a quick question (low risk)	Quick Answer Overlay	Intent bar (peek)
Analyze book / KPI / risks / opportunities	Deep Answer (Synthesis)	Sidebar
Draft email / meeting prep / talking points	Deep Answer + Artifact	Sidebar
Send/submit/approve/book/trade	Guided Action Flow	Sidebar + confirmation gates
Help inside a form (ticket/KYC)	Inline Assist	Inline + Sidebar if needed
System says "requires attention"	Proactive Card	Rail -> Sidebar/Flow